

DATABASE PROGRAMMING

Database:

A database is a software and it is an organized collection of data.

Data organized in a database in the form of tables.

Each table contains fields and records.

Database Application:

An application that communicates with database is known as database application.

Application:

An application is a program in which we interact with on the desktop.

Application is a front end & database is a back end.

Application uses SQL to communicate with database.

SQL:

SQL stands for Structured Query Language.

Database software contains two parts:

- 1) Database Application (Example: SQL*Plus)
- 2) Database (Example: Oracle)

List of databases:

- 1) Oracle
- 2) MySQL
- 3) MS-SQL Server
- 4) MS-Access
- 5) DB2

- 6) Derby
- 7) Sybase
- 8) DBase
- 9) FoxPro
- 10) Visual FoxPro .. etc.,

Application is called as client and database is called as server.

Client:

A client is a software that sends the request to server to get the response.

Server:

A server is a software that receives request from the client, process the request, constructs the response and sends the response back to a client.

Database Programming in Python:

Follow the steps to establish a connection between python application and oracle database:

- 1) Check your operating system is 32 bit or 64bit.
- 2) If your operating system is 32 bit, then install python software also 32 bit. If your operating system is 64 bit then install python software also 64bit.
- 3) Download and Install Oracle Software also according to your operating system.
- 4) Use the following at the command prompt to install cx_Oracle module:

```
C:\>pip install cx_Oracle
```

- 5) Write the following program in notepad or in IDLE

```
import cx_Oracle  
  
con=cx_Oracle.connect("system/manager@localhost:1521/x")  
  
print("Connection Established Successfully")  
  
con.close()
```

6) Run the above program

7) "system" is a username

8) "manager" is a password while installing oracle software specified.

9) "localhost" is a domain name. If the database software installed on same computer then use domain name as a localhost, otherwise use computer name.

10) "1521" is a port number. To check the port number open "tnsnames.ora" file in the following directory:

C:\oracle\app\oracle\product\11.2.0\server\network\ADMIN

11) "xe" is a service id. To get the service id type the following sql query at the SQL prompt.

```
SQL>select * from global_name;
```

12) To open sql prompt, Click on start button and type sqlplus then press enter key.

Program to establish a connection between python application and oracle database:

```
import cx_Oracle  
con=cx_Oracle.connect("system/manager@localhost:1521/xe")  
print("Connection Established Successfully")  
con.close()
```

Program to create a table:

```
import cx_Oracle  
con=cx_Oracle.connect("system/manager@localhost:1521/xe")  
cur=con.cursor()  
cur.execute("create table student(rollno number(3), name varchar2(10), marks  
number(3))")  
print("Table Created Successfully")
```

```
con.close()
```

Program to insert a record:

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/xe")
cur=con.cursor()
cur.execute("insert into student values(1, 'aaa', 99)")
con.commit()
print("One Record Inserted Successfully")
con.close()
```

Program to insert multiple records:

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/xe")
cur=con.cursor()
rows=[(2, 'bbb', 55),(3, 'ccc', 87),(4, 'ddd', 71)]
cur.executemany("insert into student(rollno, name, marks) values(:1,:2,:3)", rows)
con.commit()
con.close()
```

Program to execute dynamic sql query:

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/xe")
cur=con.cursor()
r=int(input("Enter Roll Number: "))
n=input("Enter Name: ")
m=int(input("Enter Marks: "))
```

```
cur.prepare("insert into student values(:rollno,:name,:marks)")
cur.execute(None, {"rollno":r, "name":n, "marks":m})
con.commit()
print("One Record Inserted Successfully")
con.close()
```

Program to update a record:

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/x")
cur=con.cursor()
cur.execute("update student set marks=99 where rollno=1")
con.commit()
print("One Record Updated Successfully")
con.close()
```

Program to delete a record:

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/x")
cur=con.cursor()
cur.execute("delete from student where rollno=1")
con.commit()
print("One Record Deleted Successfully")
con.close()
```

Program to insert record dynamically:

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/x")
```

```
r=int(input("Enter Roll Number: "))
n=input("Enter Name: ")
m=int(input("Enter Marks: "))
cur=con.cursor()
cur.execute("insert into student values(:rollno, :name, :marks)", [r, n, m])
con.commit()
con.close()
```

PL/SQL Procedure Example:

```
create or replace procedure insertpro(r student.rollno%type, n
student.name%type, m student.marks%type) as
begin
insert into student values(r, n, m);
end; /
```

Program to call stored procedure(PL/SQL):

```
import cx_Oracle
con=cx_Oracle.connect("system/manager@localhost:1521/xe")
r=int(input("Enter Roll Number: "))
n=input("Enter Name: ")
m=int(input("Enter Marks: "))
cur=con.cursor()
cur.callproc("insertpro", [r, n, m])
con.commit()
print("One Record Inserted Successfully")
con.close()
```